



REVIVING DUG WELLS A STEP-BY-STEP GUIDE

Restoring Community Water Security Through Proven Methods

READ THROUGH THIS ESSENTIAL GUIDE FOR LOCAL
ENGINEERS AND FIELD TEAMS

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Why Revive Dug Wells?



Karanki Village well

India, the world's largest extractor of groundwater, relies on wells for over 61% of its irrigation. Excessive extraction through deep borewells has caused rapid aquifer depletion. Reviving traditional dug wells that tap into shallow aquifers offers a sustainable, decentralized solution. In Gurugram, a total of **424 dug wells** were surveyed and identified—**71 active** with water, **331 inactive** without water, **19 cemented** or sealed, and **3 nearing disuse**—along with over **300 borewells**. By capturing rainwater, these wells have the potential to recharge groundwater, strengthen climate resilience, and ensure water availability even during droughts.

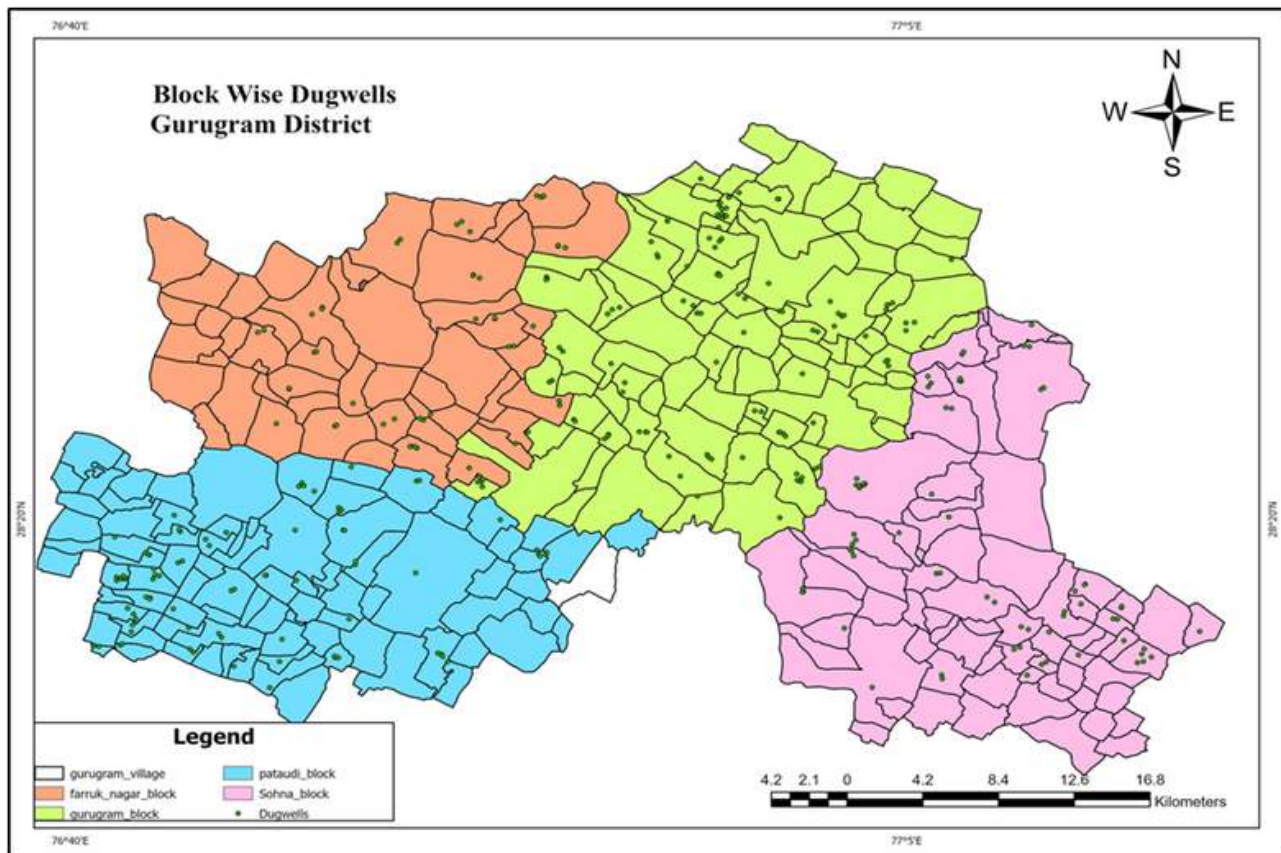
Revival of dug wells is not only cost-effective and eco-friendly—utilizing existing infrastructure—but also restores the cultural and communal value these wells once held as the heart of village life.

Why Revive Dug Wells?

This **Standard Operating Procedure (SOP)** aligns with national flagship programs. Under MGNREGA, it fulfills high-priority works like “water conservation” and “renovation of traditional water bodies.” It also advances **Atal Bhujal Yojana (ABHY)** by enabling “community-led participatory groundwater management” and supports **Jal Jeevan Mission (JJM)** through “source sustainability,” ensuring long-term water security by recharging the very aquifers that feed piped water schemes.

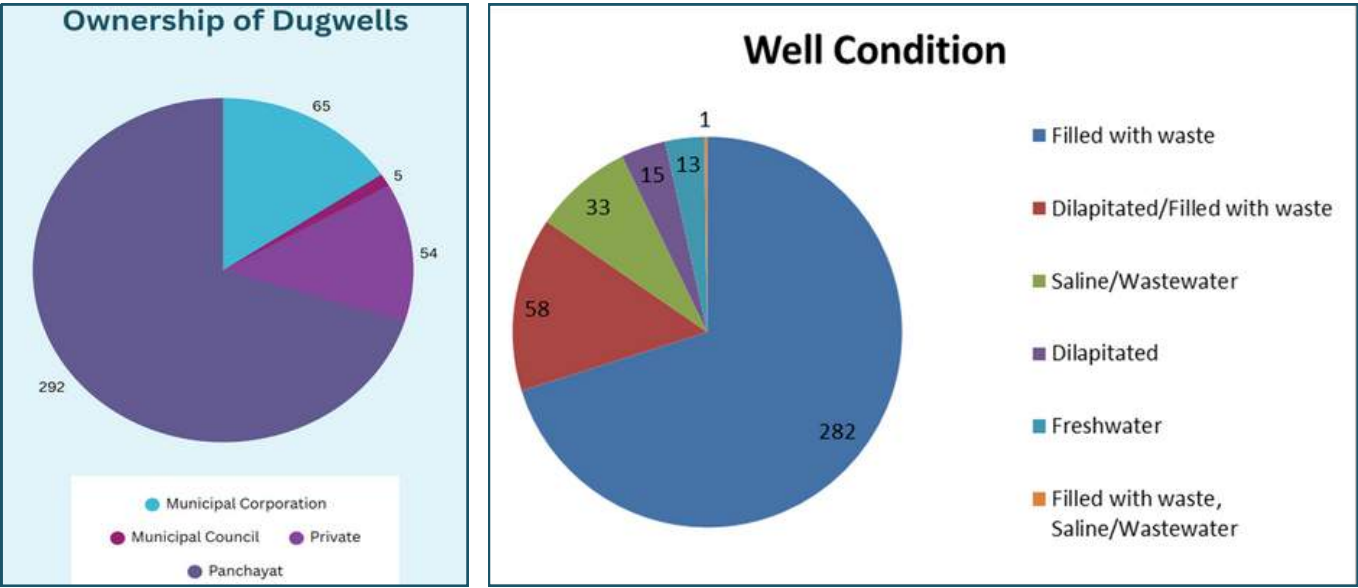


Block-wise distribution of Dug Wells surveyed

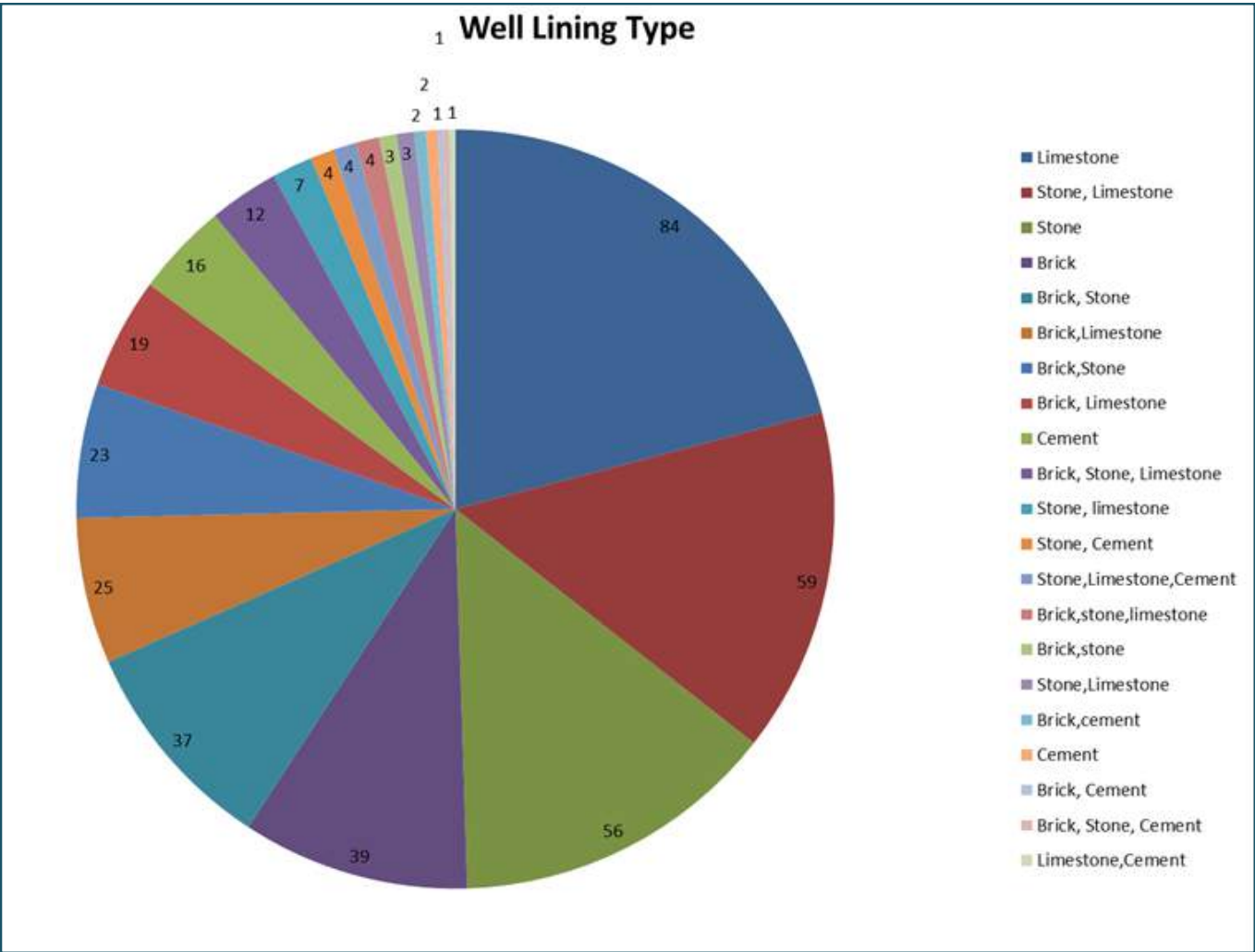


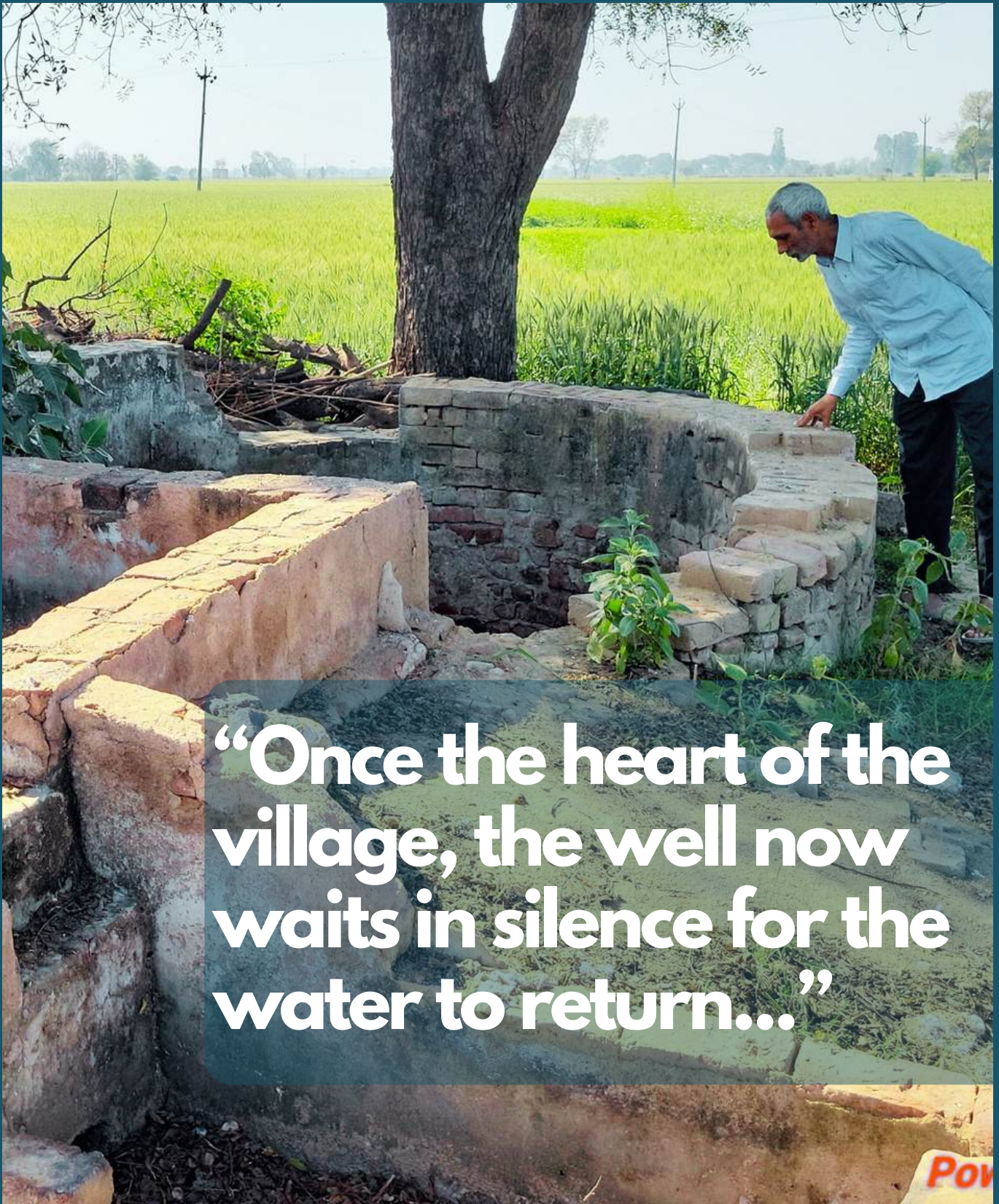
Sl. No	Name of the Block	Dug Wells
1	Gurugram	142
2	Pataudi	97
3	Farukhnagar	88
4	Sohna	97
	Total	424

Ownership and Condition of Wells



Types of Well Lining – Distribution Graph





Methodology

Standard Operating Procedure (SOP) for Dug Well Revival



Damdama Village well

1

Site Identification and
Prioritization

2

Baseline Assessment

3

Cleaning and Desilting

4

Structural Restoration

5

Filter Media and Recharge
Enhancement

6

Safety and Aesthetic
Improvements

7

Monitoring and
Maintenance

Step 1:

Site Identification and Prioritization



Khandelwla Village well

→ Assessment Criteria

Select well through pre-survey reconnaissance, satellite imagery, and community inputs, and then evaluated using standardized parameters. Selection focused on structural integrity, accessibility, recharge potential, and historical community significance. Data collection using ArcGIS for real-time entry and spatial mapping.



→ Strategic Location

Prioritise wells near public institutions such as schools and panchayat offices, or within residential clusters to ensure ease of maintenance and sustained community engagement.

Step 2:

Baseline Assessment

→ PHASE 01

Technical Inspection

Measure well depth, diameter, current water level, and silt accumulation. Document structural conditions and identify repair needs.

→ PHASE 02

Water Quality Testing

Test for pH, Total Dissolved Solids (TDS), nitrate levels, and coliform bacteria to establish baseline water safety conditions and assess the suitability of well water for domestic and agricultural use through a NABL-accredited laboratory, following CPCB standard drinking water protocols.

→ PHASE 03

Historical Documentation

Record previous usage patterns, seasonal variations, and cultural importance to the community for future reference and planning.

Step 3:

Cleaning and Desilting



Daultabad Village well



Cleaning of Daultabad Village well

→ PHASE 01

Removal Process

Remove accumulated waste and sludge using manual or mechanical methods appropriate to well size and condition.

→ PHASE 02

Waste Disposal

Dispose of solid waste responsibly through local municipal systems following environmental guidelines.

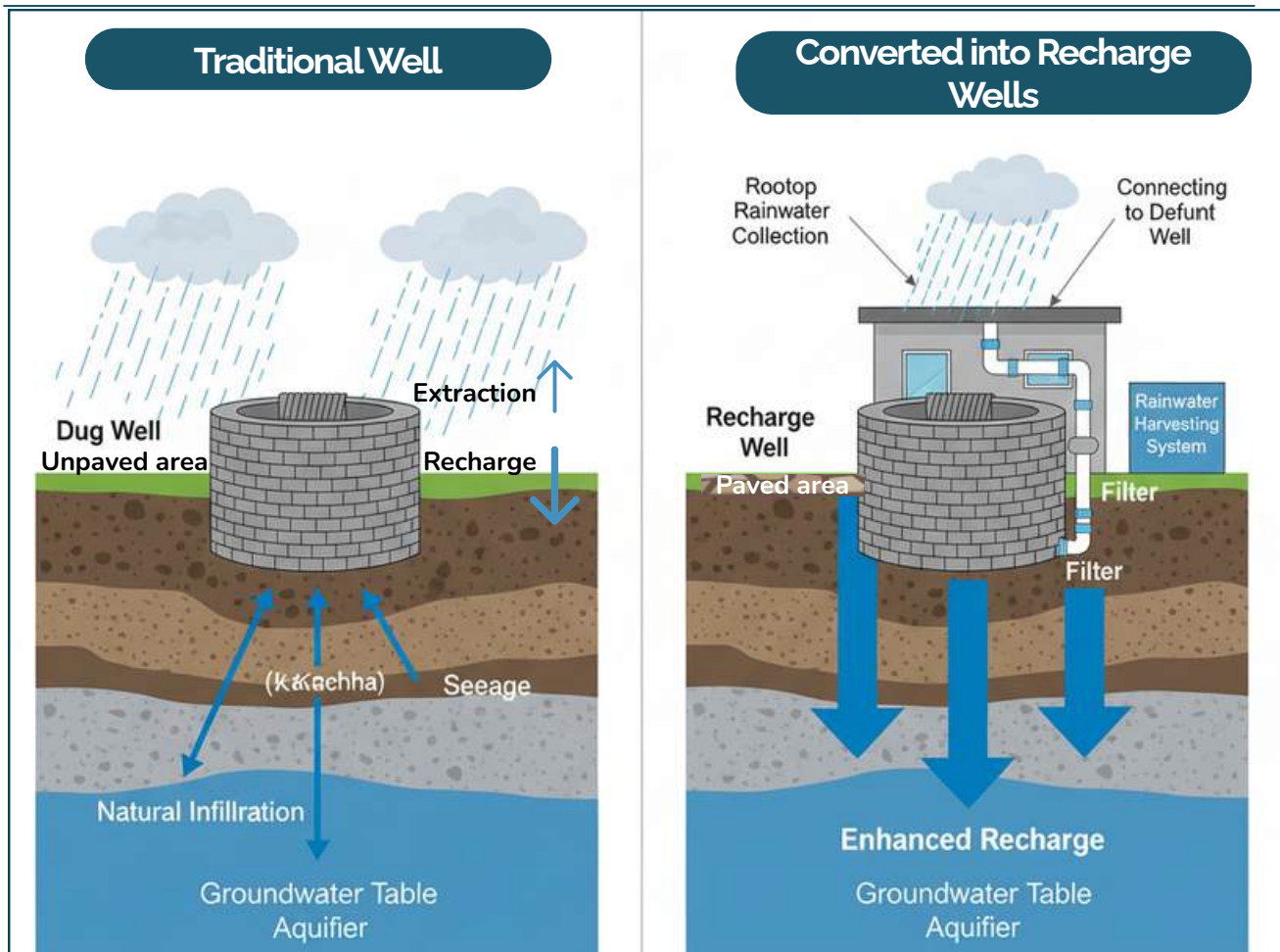
→ PHASE 03

Water Management

Pump out stagnant water to enable thorough inspection and facilitate deep cleaning of well walls and base.

Step 4:

Structural Restoration



→ Parapet and Lining Repairs

Repair or rebuild parapet walls, internal linings, and retaining structures using locally sourced materials that match traditional construction methods.

→ Contamination Prevention

Strengthen inlet and outlet points. Seal all cracks and fissures to prevent surface water contamination and maintain structural integrity.

Step 5:

Filter Media and Recharge Enhancement



Connecting the pipe to the well to enhance well recharge.

Graded Filter Media

Fill the well base with layers of coarse aggregates, sand, and charcoal to create an effective natural filtration system.

Rainwater Connectivity

Connect rooftop rainwater pipes and stormwater channels to the well through properly designed filtration chambers.

Maintenance Infrastructure

Install silt traps and inspection chambers for easy access during routine maintenance and monsoon preparation activities.

Step 6:

Safety and Aesthetic Improvements

→ PHASE 01

Safety Infrastructure

Install secure fencing or elevated parapet walls around the well perimeter to prevent accidents and ensure public safety, especially for children.

→ PHASE 02

Educational Signage

Add weatherproof signs detailing well history, recharge function, and usage guidelines to educate the community and visitors.

→ PHASE 03

Landscape Enhancement

Where appropriate, improve surrounding areas with native plants and paving to enhance visibility, accessibility, and foster local ownership and pride.

Step 7:

Monitoring and Maintenance

→ PHASE 01

Establish Local Committee

Form a monitoring committee including school representatives, panchayat members, and citizen volunteers to oversee ongoing well management.

→ PHASE 02

Bi-Annual Desilting

Conduct desilting operations twice yearly to remove accumulated sediment and maintain optimal water storage capacity throughout all seasons.

→ PHASE 03

Annual Quality Testing

Perform comprehensive water quality testing once per year to ensure safety standards are maintained and identify any emerging contamination issues.

→ PHASE 04

Pre-Monsoon Preparation

Clean rainwater inlet channels and service filter chambers before monsoon season to maximise recharge potential and prevent blockages during heavy rainfall.

Well of Khandewla Village

Before



29th May 2025



After

24th Sept 2025

Well of Daultabad Village

Before



18th May 2025



After

24th October 2025

Key Success Factors

Community Ownership

Engage local residents from initial assessment through ongoing maintenance to ensure sustained commitment and proper care of restored wells.



Regular Maintenance

Establish clear schedules for desilting, testing, and seasonal preparation to prevent deterioration and maintain water quality standards consistently.

Quality Monitoring

Implement systematic testing protocols and document results to track improvements and identify any issues requiring immediate attention or intervention.



Our Citizen Scientist

The well inventory survey in Gurugram was strengthened by the efforts of four dedicated citizen scientists—local residents who transformed from community members into field investigators through focused training and mentorship. Coming from different villages across the district, they had firsthand experience of the region's growing water challenges. Recognizing their interest and familiarity with local water sources, they were trained by the GuruJal team in scientific data collection, field measurement techniques, and the use of standardized survey tools.

Over a period of 45 days, these citizen scientists travelled across more than 200 villages, documenting 424 dug wells and over 300 borewells. Equipped with simple field tools and a deep sense of purpose, they measured well dimensions, recorded water levels, and engaged with local residents to capture stories of changing groundwater conditions. Their involvement not only ensured accurate and extensive data collection but also built a strong bridge between science and society. What began as a training exercise evolved into a journey of empowerment—turning local villagers into stewards of their community's most precious resource: water.



Ajay

Citizen Scientist: Sohna Block



Sandeep

Citizen Scientist: Pataudi Block



Virendra

Citizen Scientist: Gurugram Block



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ude: 77.095185
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acy: 3.1 m
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Annexure

(Water Quality Test Report)



Envirochem Testing & Consultancy LLP

(Formerly known as Envirochem Testing Lab & Research Centre)

(GOVT. APPROVED LAB)

(An ISO 9001 : 2015, ISO 14001 : 2015, ISO 45001 : 2018 Certified Lab)

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TEST REPORT

Report No & ULR No.	ETL/PNP/251615 & TC601525000003927F	Report Date	24.06.2025	Doe No.	ETL/QF/7.8/01
Issue to: M/s Gurujal-Abhipsa Foundation F- 6/9, F-2, Block-F, DLF Phase-1, Sector 26A, Gurugram		Party's Ref No: Nil Work Order No: 2506/1615 Period of Testing: 17.06.2025-24.06.2025			
SAMPLE PARTICULARS					
1	Name of site	:	M/s Gurujal-Abhipsa Foundation		
2	Type of sample	:	Ground Water (Borewell)		
3	Point of Sample Collection	:	FarrukNahar Block - Borewell Near Pond (Khandewla)		
4	Date of sample collection/received	:	16.06.2025		
5	Purpose of analysis	:	Monitoring purpose		
6	Sample collected/supplied by	:	By Lab Representative		
7	Quantity of Sample	:	5 Litre + 250 ml		
8	Method of Sampling	:	IS3025(P-1)1987		

TEST RESULTS

Sr. No.	Parameters	Results	Protocol used
1.	pH	7.42	IS 3025 (P-11) 2022
2.	Electrical Conductivity, $\mu\text{mhos/cm}$	1792	IS 3025 (P-14) 1984
3.	Total Dissolved Solids, mg/l	1252	IS 3025 (P-16) 2023
4.	Total Suspended Solids, mg/l	<1	IS 3025 (P-17) 2022
5.	Total Hardness (as CaCO_3), mg/l	470	IS 3025 (P-21) 2009
6.	Chloride (as Cl), mg/l	352	IS 3025 (P-32) 1988
7.	COD, mg/l	<5	IS 3025 (P-58) 2023
8.	BOD at 27°C for 3 Days, mg/l	<2	IS 3025 (P-44) 2022
9.	Ammonical Nitrogen (as N), mg/l	0.86	IS 3025 (P-34) 2023
10.	Total Nitrogen (as N), mg/l	4.92	IS 3025 (P-34) 2023
11.	Total Residual Chlorine, mg/l	<0.1	IS 3025 (P-26) 2021
12.	Nitrate Nitrogen, mg/l	3.3	IS 3025 (P-34) 2023
13.	Sulphide (as S), mg/l	<0.05	IS 3025 (P-29) 2022
14.	Total Phosphate (as P), mg/l	0.43	IS 3025(P-31) 2021
15.	Dissolved Phosphate (as PO_4), mg/l	0.91	IS 3025 (P-31) 2021
16.	Flouride(as F), mg/l	0.23	APHA 24 th Edition (P-4500) 2023
17.	Boron (as B), mg/l	<0.02	IS 3025 (P-57) 2021
18.	Sulphate (as SO_4), mg/l	218	IS 3025 (P-24) 2022

Remarks : 1. All Parameters are covered under NABL Scope.

*****End Report*****

[Signature]

Reviewed By
Chemist

[Signature]

Verified By
Neha Singh (T.M)



Authorised By
Dr. Rajender Kumar

- N 1. Samples shall be disposed off after 15 days issue of test report unless specified.
O 2. Results listed above related to the tested samples, Endorsement of the same is neither inferred nor implemented.
T 3. The test report shall not be reproduced full or in part & can't be used as proof in the court of law.
E 4. The test report should not be used in any advertising agency/media without the written approval of laboratory



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Issue to: M/s Gurujal-Abhipsa Foundation F- 6/9, F-2, Block-F, DLF Phase-I, Sector 26A, Gurugram		Party's Ref No: Nil Work Order No: 2506/1615 (A) Period of Testing: 17.06.2025-24.06.2025			

SAMPLEPARTICULARS

1	Name of site	:	M/s Gurujal-Abhipsa Foundation
2	Type of sample	:	Ground Water (Borewell)
3	Point of Sample Collection	:	FarrukNahar Block - Borewell Near Pond (Khandewla)
4	Date of sample collection/received	:	16.06.2025
5	Purpose of analysis	:	Monitoring purpose
6	Sample collected/supplied by	:	By Lab Representative
7	Quantity of Sample	:	5 Litre + 250 ml
8	Method of Sampling	:	IS3025(P-1)1987

TEST RESULTS

Sr. No.	Parameters	Results	Protocol used
1.	Sodium Absorption Ratio (SAR)	7.8	APHA Method
2.	Residual Sodium Carbonate (as RSC), mg/l	252	IS 11624-1986
3.	Anionic Surfactants (MBAS), µg/l	ND	IS 13428 (Annex-K)
Bacteriological Examinations			
1.	Fecal Coliform MPN/100 ml	Absent	IS 1622 : 1981
2.	E. Coli	Absent	IS 1622 : 1981

Remarks: 1. All Parameters are not covered under NABL Scope. 2. ND- Not Detectable.

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Issue to: M/s Gurujal-Abhipsa Foundation F- 6/9, F-2, Block-F, DLF Phase-I, Sector 26A, Gurugram		Party's Ref No: Nil Work Order No: 2506/1616 Period of Testing: 17.06.2025-24.06.2025			

SAMPLEPARTICULARS

1	Name of site	:	M/s Gurujal-Abhipsa Foundation
2	Type of sample	:	Ground Water (Borewell)
3	Point of Sample Collection	:	FarrukNahar Block - Borewell house no. 259 (Khandewala)
4	Date of sample collection/received	:	16.06.2025
5	Purpose of analysis	:	Monitoring purpose
6	Sample collected/supplied by	:	By Lab Representative
7	Quantity of Sample	:	5 Litre + 250 ml
8	Method of Sampling	:	IS3025(P-1)1987

TEST RESULTS

Sr. No.	Parameters	Results	Protocol used
1.	pH	7.80	IS 3025 (P-11) 2022
2.	Electrical Conductivity, $\mu\text{mhos/cm}$	1943	IS 3025 (P-14) 1984
3.	Total Dissolved Solids, mg/l	1360	IS 3025 (P-16) 2023
4.	Total Suspended Solids, mg/l	<1	IS 3025 (P-17) 2022
5.	Total Hardness (as CaCO_3), mg/l	489	IS 3025 (P-21) 2009
6.	Chloride (as Cl), mg/l	384	IS 3025 (P-32) 1988
7.	COD, mg/l	<5	IS 3025 (P-58) 2023
8.	BOD at 27°C for 3 Days, mg/l	<1	IS 3025 (P-44) 2022
9.	Ammonical Nitrogen (as N), mg/l	0.34	IS 3025 (P-34) 2023
10.	Total Nitrogen (as N), mg/l	3.8	IS 3025 (P-34) 2023
11.	Total Residual Chlorine, mg/l	<0.1	IS 3025 (P-26) 2021
12.	Nitrate Nitrogen, mg/l	2.9	IS 3025 (P-34) 2023
13.	Sulphide (as S), mg/l	<0.05	IS 3025 (P-29) 2022
14.	Total Phosphate (as P), mg/l	0.22	IS 3025(P-31) 2021
15.	Dissolved Phosphate (as PO_4), mg/l	0.46	IS 3025 (P-31) 2021
16.	Flouride(as F), mg/l	0.27	APHA 24 th Edition (P-4500) 2023
17.	Boron (as B), mg/l	<0.02	IS 3025 (P-57) 2021
18.	Sulphate (as SO_4), mg/l	258	IS 3025 (P-24) 2022

Remarks : I. All Parameters are covered under NABL Scope.

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Issue to: M/s Gurujal-Abhipsa Foundation F- 6/9, F-2, Block-F, DLF Phase-1, Sector 26A, Gurugram		Party's Ref No: Nil Work Order No: 2506/1616 (A) Period of Testing: 17.06.2025-24.06.2025			

SAMPLEPARTICULARS

1	Name of site	:	M/s Gurujal-Abhipsa Foundation
2	Type of sample	:	Ground Water (Borewell)
3	Point of Sample Collection	:	FarrukNahar Block - Borewell house no. 259 (Khandewala)
4	Date of sample collection/received	:	16.06.2025
5	Purpose of analysis	:	Monitoring purpose
6	Sample collected/supplied by	:	By Lab Representative
7	Quantity of Sample	:	5 Litre + 250 ml
8	Method of Sampling	:	IS3025(P-1)1987

TEST RESULTS

Sr. No.	Parameters	Results	Protocol used
1.	Sodium Absorption Ratio (SAR)	8.1	APHA Method
2.	Residual Sodium Carbonate (as RSC), mg/l	313	IS 11624-1986
3.	Anionic Surfactants (MBAS), µg/l	ND	IS 13428 (Annex-K)
Bacteriological Examinations			
1.	Fecal Coliform MPN/100 ml	Absent	IS 1622 : 1981
2.	E. Coli	Absent	IS 1622 : 1981

Remarks: 1. All Parameters are not covered under NABL Scope. 2. ND- Not Detectable.

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